

WEEK 1

Task1

Develop a C program that demonstrates how a Linux operating system executes a command entered by a user that

1. Accept a Linux command as input.
2. Create a child process using fork().
3. Execute the command in the child process using an appropriate exec() system call.
4. Allow the parent process to wait for the child using wait ().
5. Display the Process ID (PID) of both parent and child processes.

Source Code:

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>
#include <string.h>

int main(){
    char command[100];
    pid_t pid;
    printf("Enter a Linux command: ");
    scanf("%s", command);
    pid = fork();
    if(pid < 0) {
        printf("Fork failed!\n");
        return 1;
    }
    else if(pid == 0) {
        printf("\nChild Process\n");
        printf("Child PID : %d\n", getpid());

        execlp(command, command, NULL);
        printf("Command execution failed.\n");
        exit(1);
    } else {
        wait(NULL);
        printf("\nParent Process\n");
        printf("Parent PID : %d\n", getpid());
        printf("Child process completed.\n");
    }
    return 0;
}
```

Output:

```
Enter a Linux command: ls

Child Process
Child PID : 437
OSSP code.c files.c gm.c hello.c os program.c programming.c source_code
code files gm hello operating os.c programming programming.c.save source_code.c

Parent Process
Parent PID : 436
Child process completed.
```

Task2

Using Linux terminal commands (uname, lscpu, lsblk, ps, top), investigate the relationship between hardware resources and operating system services. Prepare a report explaining how the OS abstracts CPU, memory, storage, and I/O devices.

Report

1. Uname: Displays operating system and kernel information.

```
balaji@BalajiPatil:~$ uname
Linux
balaji@BalajiPatil:~$ |
```

2. lscpu: Displays CPU information.

3. lsblk: Displays storage devices.

```
balaji@BalajiPatil:~$ lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda         8:0    0 356.9M  1 disk
sdb         8:16    0 159.4M  1 disk
sdc         8:32    0    2G    0 disk [SWAP]
sdd         8:48    0    1T    0 disk /snap
                                     /mnt/wslg/distro
                                     /

balaji@BalajiPatil:~$ |
```

4. ps: Displays currently running processes.

```
balaji@BalajiPatil:~$ pr
      PIN  TTY          TIME CMD
      322  pts/0        00:00:00 bash
      526  pts/1        00:00:00 pr
balaji@BalajiPatil:~$ |
```

5. top: Displays real-time system performance.

balaji@BalajiPatil:~\$ top

top - 09:40:06 up 4 min, 1 user, load average: 0.06, 0.08, 0.03

Tasks: 29 total, 1 running, 28 sleeping, 0 stopped, 0 zombie

%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni, 99.9 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st

MiB Mem : 7794.6 total, 6468.8 free, 542.7 used, 934.9 buff/cache

MiB Swap: 2048.0 total, 2048.0 free, 0.0 used, 7252.0 avail Mem

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+
1	root	20	0	21628	13156	9796	S	0.0	0.2	0:00.41
2	root	20	0	3180	2212	2072	S	0.0	0.0	0:00.01
6	root	20	0	3196	2132	2008	S	0.0	0.0	0:00.00
52	root	19	-1	42164	16232	15156	S	0.0	0.2	0:00.09
99	root	20	0	25336	6736	5124	S	0.0	0.1	0:00.12
111	root	20	0	152944	1768	1512	S	0.0	0.0	0:00.00
114	root	20	0	153076	1656	1396	S	0.0	0.0	0:00.00
116	root	20	0	153076	1828	1512	S	0.0	0.0	0:00.00
125	root	20	0	153076	1704	1444	S	0.0	0.0	0:00.00
130	root	20	0	153076	1680	1424	S	0.0	0.0	0:00.00
135	root	20	0	526764	12536	1536	S	0.0	0.2	0:01.90
160	systemd+	20	0	21460	13192	10932	S	0.0	0.2	0:00.06
162	systemd+	20	0	91028	7944	6976	S	0.0	0.1	0:00.04
217	root	20	0	4244	2684	2432	S	0.0	0.0	0:00.00
218	message+	20	0	9632	5304	4600	S	0.0	0.1	0:00.03
227	root	20	0	2147016	42272	24680	S	0.0	0.5	0:00.37
231	root	20	0	17964	8852	7812	S	0.0	0.1	0:00.04
233	root	20	0	1756104	12996	10888	S	0.0	0.2	0:00.07
247	syslog	20	0	222516	5976	4572	S	0.0	0.1	0:00.04
256	root	20	0	3124	1976	1836	S	0.0	0.0	0:00.00
270	root	20	0	107020	23188	13728	S	0.0	0.3	0:00.08
380	root	20	0	3196	1120	980	S	0.0	0.0	0:00.00
381	root	20	0	3196	1252	1108	S	0.0	0.0	0:00.01
382	balaji	20	0	7568	6144	4372	S	0.0	0.1	0:00.03
383	root	20	0	6704	4648	3864	S	0.0	0.1	0:00.00
427	balaji	20	0	20376	11652	9492	S	0.0	0.1	0:00.06
428	balaji	20	0	21152	3576	1852	S	0.0	0.0	0:00.00
454	balaji	20	0	6080	5360	3692	S	0.0	0.1	0:00.01
629	balaji	20	0	9300	5760	3572	R	0.0	0.1	0:00.01